

Quantum Neuromorphic Architecture with Memristive Synapses:

A Theoretical Model, Analysis of Fundamental Limitations,
and Prospects for Physical Implementation

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Abstract

This paper presents a theoretical model of a computational architecture that unifies principles of quantum computing, neuromorphic systems, and memristive technologies. Qubits serve as neurons, while quantum memristors act as programmable synaptic connections whose state modulates the effective coupling constant between computational elements. We formulate the mathematical apparatus based on the Lindblad master equation for open quantum systems, analyze fundamental limitations of quantum learning (no-cloning theorem, barren plateaus, Landauer’s principle), and evaluate the feasibility of physical implementation using current experimental data. This work constitutes a theoretical framework for future research, not a ready-made engineering solution.

Keywords: quantum computing, neuromorphic systems, quantum memristor, variational quantum algorithms, decoherence, barren plateaus, Landauer’s principle.

1 Introduction

Modern computing is approaching fundamental limits. Moore’s Law is losing relevance, and the von Neumann architecture creates a bottleneck between memory and processor. Three promising directions are developing in parallel: quantum computing, neuromorphic systems, and memristive technologies.

Quantum computing offers exponential speedup for specific tasks but faces decoherence and scalability challenges. Neuromorphic systems demonstrate energy efficiency for AI tasks but are limited in universality. Memristive technologies enable in-memory computing but face fundamental limitations in classical implementations.

The concept of the ideal memristor has been critically examined: it has been shown that the dynamic state equations for the HP memristor violate Landauer’s principle regarding the minimum energy required to change information states [14]. Moreover, memristors controlled solely by current or voltage cannot protect their memory states from Johnson–Nyquist noise and will continuously lose information.

Integrating these approaches represents a strategic challenge that requires accounting for the fundamental limitations of each. Recent reviews on quantum neuromorphic computing emphasize that “fully autonomous quantum learning remains theoretical” [11].

2 Theoretical Model

2.1 Architectural Paradigm

The computational system implements a directed graph $G(V, E)$, where $V = \{v_i\}$ is the set of qubit-neurons and $E = \{e_{jk}\}$ is the set of quantum memristor-synapses.

Each qubit-neuron is described by a state vector:

$$|\psi_i\rangle = \alpha_i|0\rangle + \beta_i|1\rangle, \quad |\alpha_i|^2 + |\beta_i|^2 = 1. \quad (1)$$

Definition of a quantum memristor. In the current literature, a quantum memristor is defined as a quantum device exhibiting nonlinear, history-dependent behavior analogous to a classical memristor, but with genuinely quantum input and output signals. Experimentally realized implementations include:

- (1) **Optical quantum memristor** based on vacuum–single-photon qubits passing through a tunable beam splitter whose reflectivity adapts to the mean photon number in the device, preserving input state coherence [1].
- (2) **Quantum conductance memristors** exhibiting conductance quantization in units of $G_0 = 2e^2/h \approx 77.5 \mu\text{S}$. Ta_2O_5 -based devices have demonstrated 6-bit memory with 68 conductance states [2].
- (3) **MoS₂-based memristors** with atomic-scale parameter confinement, showing reproducible quantum conductance states from $1G_0$ to $4G_0$ and light-induced half-integer states [3].

Contrary to a common simplification, a quantum memristor does not exist in a “superposition of conductance states.” Real implementations use either classical feedback (optical memristor), hysteretic polarization behavior, or conductance quantization.

2.2 System Dynamics

The evolution of the system is described by the Lindblad master equation for the density matrix, which provides a correct description of an open quantum system:

$$\frac{d\rho}{dt} = -\frac{i}{\hbar}[\hat{H}_{\text{total}}, \rho] + \sum_k \gamma_k \left(L_k \rho L_k^\dagger - \frac{1}{2} \{L_k^\dagger L_k, \rho\} \right), \quad (2)$$

where γ_k are relaxation rates and L_k are Lindblad operators.

The total Hamiltonian:

$$\hat{H}_{\text{total}} = \hat{H}_{\text{neurons}} + \hat{H}_{\text{synapses}} + \hat{H}_{\text{interaction}} + \hat{H}_{\text{control}}. \quad (3)$$

Neuron self-energy:

$$\hat{H}_{\text{neurons}} = \sum_i (\varepsilon_i \sigma_z^i + \Delta_i \sigma_x^i). \quad (4)$$

Interaction:

$$\hat{H}_{\text{interaction}} = \sum_{jk} J_{jk}(t) \sigma_j^\alpha \otimes \sigma_k^\beta, \quad (5)$$

where $J_{jk}(t) = \langle \varphi_{jk}(t) | \hat{G}_{jk} | \varphi_{jk}(t) \rangle$ is the effective coupling constant controlled by the memristor state.

The use of the Lindblad equation rather than the Schrödinger equation is essential, as quantum systems inevitably interact with the environment. Unitary evolution cannot describe learning processes that require irreversibility.

2.3 Learning Mechanism

Learning is implemented through Variational Quantum Algorithms (VQAs). Consider a loss function $L(\theta)$, where $\theta = \{\theta_{jk}\}$ are the memristor parameters.

The parameter shift rule—the standard method for gradient estimation:

$$\frac{\partial L}{\partial \theta_{jk}} = \frac{L(\theta_{jk} + \pi/2) - L(\theta_{jk} - \pi/2)}{2}. \quad (6)$$

Fundamental limitations:

1. **No-cloning theorem.** The no-cloning theorem makes standard backpropagation impossible, as intermediate quantum states cannot be copied for reuse. This leads to prohibitive quantum resource costs when applying VQAs to large-scale tasks [4].
2. **Barren plateaus.** Gradients in variational quantum circuits decay exponentially with the number of qubits, fundamentally limiting the scalability of VQAs [15].
3. **Computational cost.** The parameter shift rule requires $O(M)$ circuit evaluations for M parameters per iteration. The computational complexity of gradient estimation grows linearly with the number of parameters [6].

Memristor state updates are implemented through classical or semi-classical control pulses that modify memristor parameters.

3 Stability and Performance Analysis

3.1 Decoherence Resistance

Stability analysis uses the Lindblad equation. Specific relaxation rates γ_k depend on the platform. Experimental data for various platforms are summarized in Table 1.

Conclusion: Realistic coherence times for silicon platforms range from hundreds of nanoseconds to several milliseconds. For MATBG qubits, experimental data are absent.

Table 1: Coherence times for various qubit platforms.

Platform	Coherence time	Conditions
Si-MOS spin qubits	$T_2^* = 143$ ns, $T_2^{\text{Hahn}} = 40.7$ μ s	Phase modulation [7]
$^{28}\text{Si}/\text{SiGe}$ spin qubits	$T_2^* < 1$ μ s	Weak magnetic fields [8]
Si spin qubits (300 mm fab)	$T_2^{\text{Hahn}} = 4$ ms	Singlet–triplet oscillations [9]
Graphene qubits (MATBG)	Theoretically protected	Not yet experimentally realized [10]

3.2 Performance Estimation

Energy efficiency:

- Energy per quantum operation: $E_{\text{op}} \approx \hbar\omega_0 \approx 10^{-24}$ J at $\omega_0 \approx 10^{10}$ Hz.
- Landauer’s principle: the minimum energy for irreversible bit erasure at $T = 4$ K is $E = k_B T \ln 2 \approx 4 \times 10^{-23}$ J. Landauer’s principle does not grant quantum computing an inherent advantage over classical computing: erasing one bit of information costs $kT \ln 2$ regardless of whether the bit is classical or quantum [12].
- Cryogenic cooling energy and control energy must also be accounted for.

Element density:

- Theoretical density for 100×100 nm qubits: 10^{10} elements/cm².
- Realistic density is limited by control line density, crosstalk, and heat dissipation.

Training speed:

- Parameter shift rule: $O(M)$ gradient evaluations per iteration.
- Barren plateaus: gradients decay exponentially with system size.

4 Physical Implementation

4.1 Two-Dimensional Materials Platform

Magic-angle twisted bilayer graphene (MATBG):

- At the “magic angle” $\sim 1.1^\circ$, flat bands emerge.
- $0\text{--}\pi$ qubits based on MATBG, protected from local perturbations, have been theoretically proposed. Gate-defined Josephson junctions based on MATBG have recently been fabricated [10].
- Experimentally: MATBG qubits have not yet been created.

Ferroelectric memristors based on CuInP_2S_6 (CIPS):

- CIPS exhibits dual polar and ionic character.
- Memristive behavior is driven by Cu^+ ion migration and water molecule desorption—a classical process.
- CIPS operates at room temperature but is not a quantum-coherent platform.

4.2 Control Scheme

- **Initialization:** electrostatic gate control.
- **Evolution:** microwave pulses.
- **Readout:** RF reflectometry—standard for superconducting qubits, but its application to MATBG qubits has not been demonstrated.

5 Comparative Analysis with Existing Technologies

Table 2: Comparison of the proposed architecture with existing technologies.

Parameter	Classical CPU	Quantum processor	Neuro-morphic	Proposed architecture
Energy per op.	10^{-15} J	10^{-18} J*	10^{-15} J	$\sim 10^{-24}$ J**
Element density	10^{10} cm $^{-2}$	10^6 cm $^{-2}$	10^8 cm $^{-2}$	10^{10} cm $^{-2}$ (theor.)
Universality	Full	Limited	Specialized	Theoretically full
Training speed	$O(N^2)$	$O(M)/\text{iter}$	$O(N)$	$O(M)/\text{iter}$
Coherence time	N/A	100–500 μs	N/A	143 ns – 4 ms (Si)

*Including cryogenics. **Single quantum energy; with Landauer and cooling, no advantage over existing quantum systems.

6 Conclusion and Future Directions

We have presented a theoretical model of a quantum neuromorphic architecture with memristive synapses. Key results:

1. Developed a formalism for describing the dynamics of an open quantum system with memristive connections based on the Lindblad equation.
2. Analyzed fundamental limitations of quantum learning: the no-cloning theorem makes standard backpropagation impossible and leads to prohibitive quantum resource costs [4]; barren plateaus fundamentally limit VQA scalability [15]; Landauer’s principle does not grant quantum computing an inherent energy efficiency advantage [12].
3. Reviewed experimental implementations of quantum memristors: optical (vacuum–single-photon qubits with preserved coherence [1]) and conductance-based (Ta_2O_5 devices with 6-bit, 68-state memory [2]).
4. Assessed physical platform feasibility: silicon spin qubits demonstrate coherence times from 143 ns to 4 ms [7, 9]; MATBG qubits are theoretically proposed but experimentally unrealized [10].

Promising directions:

- Development of barren plateau mitigation methods for large-scale systems.

- Experimental demonstration of elementary cells with quantum memristors.
- Investigation of hybrid platforms combining classical memristors with quantum processors.

The proposed architecture represents a theoretical framework, not a ready-made engineering solution. Practical implementation requires solving fundamental problems: quantum state stabilization, scalable training methods, and materials with controlled decoherence.

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